THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE TERMINAL EXAMINATION
031 PHYSICS
MARKING SCHEME MAY 2025

SECTION A (15 MARKS)

1. **@01 Mark = 10 marks**

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	D	B	A	C	A	B	B	A	C

2. **@01 Mark = 05 marks**

i	ii	Iii	iv	v
C	A	D	E	B

SECTION B (MARKS 70)
Answer all questions in this section

3. (a) Problem identification: Is a first step in scientific investigation, or Problem identification is the process of recognizing and clearly defining a specific challenge or question that needs to be addressed through scientific investigation or experimentation 04 marks

(b)

Observe patterns in the ball's behavior, such as where it consistently stops. Consider variables like ramp angle, friction, or ball surface. The problem might be due to surface resistance or lack of sufficient force to keep the ball moving. 06marks

4. (a) A hypothesis is a testable prediction or educated guess about the relationship between two or more variables, based on observations and prior knowledge Or is an intelligent guess 04 marks

(b) If a balloon is rubbed against hair, then it will become negatively charged and attract small, neutral objects like paper due to electrostatic force. 06marks

5. (a) A theory is a well-substantiated explanation of an aspect of the natural world that is based on a body of evidence and repeatedly tested and confirmed through observation and experimentation. 02 marks Or is a big idea or explanation that scientists come up with to describe how something in behave

(b) Explain four significance of theory of gravity

i. It explains the falling of objects to the ground. 02 marks @
ii. It helps understand planetary orbits and motion in space.
iii. It is crucial in engineering for designing stable structures.
iv. It explains tidal forces caused by the moon's gravity on Earth

6. (a) A scientific law is a statement based on repeated experimental observations that describes a natural phenomenon and predicts outcomes under specific conditions. 02marks

(b)

i. Hooke's law: The extension of a spring is directly proportional to the force applied, provided the elastic limit is not exceeded. 02marks

ii. The law of flotation: A floating body displaces its own weight of the fluid in which it floats. 02marks

iii. Pascal's Principle: it states that Pressure applied at any point to a confined fluid is transmitted equally in all directions throughout the fluid.

7.(a) To calculate the accurate diameter of the rod using the screw gauge, we need to consider the contribution of both the sleeve and the thimble, taking into the account the least count of the instrument 04 marks

Sleeve shows the main scale reading in mm.

Thimble shows finer readings in hundredths of a mm.

Least count (0.01 mm) ensures high precision.

To ensure accuracy, avoid zero error, calibrate properly, and take multiple readings.

(b)

Sleeve reading = 1.6 cm = 16.00 mm 01marks

Thimble reading = 48 × 0.01 mm = 0.48 mm 02marks

Total diameter = 16.00 + 0.48 = 16.48 mm 02 marks

So the correct reading is 16.48mm 01marks

8.(a) Force is a physical quantity that causes an object to move, stop, or change direction; it is a push or pull on an object. 02marks

(b) List four naturally occurring forces

i. Gravitational force ……02marks

ii. Electrostatic force ……02marks

iii. strong force ……02marks

iv. weak force ……02marks

9.Four differences between mass and weight

Mass	Weight
i. Measure of matter	i. Measure of gravitational pull
ii. Constant everywhere	ii. Changes with location (gravity)
iii. Measured in kilograms (kg)	iii. Measured in newton's (N)
iv. Scalar quantity	iv. Vector quantity
v. Measured by balance	v. Measured by spring balance

2marks @

SECTION C (15 MARKS)

10. (a)

i. Magnetic force**……02 marks**

ii. Frictional force **……02 marks**

iii. Gravitational force ……..**02 marks**

(b) Branches of physics

i. Mechanics – Studies motion, forces, and energy of physical objects.

ii. Thermodynamics – Deals with heat, temperature, and energy transformation.

iii. Optics – Focuses on the behavior and properties of light.

iv. Electromagnetism – Studies electric and magnetic fields and their interactions.

v. Waves and Vibrations – Explores oscillatory motion and wave phenomena such as sound.

vi. Nuclear Physics – Concerned with the structure and behavior of atomic nuclei and nuclear energy.

1.5 marks @ total 9 marks